

Health risk assessment using Monte-Carlo simulations due to arsenic contamination in groundwater in Punjab

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ABSTRACT

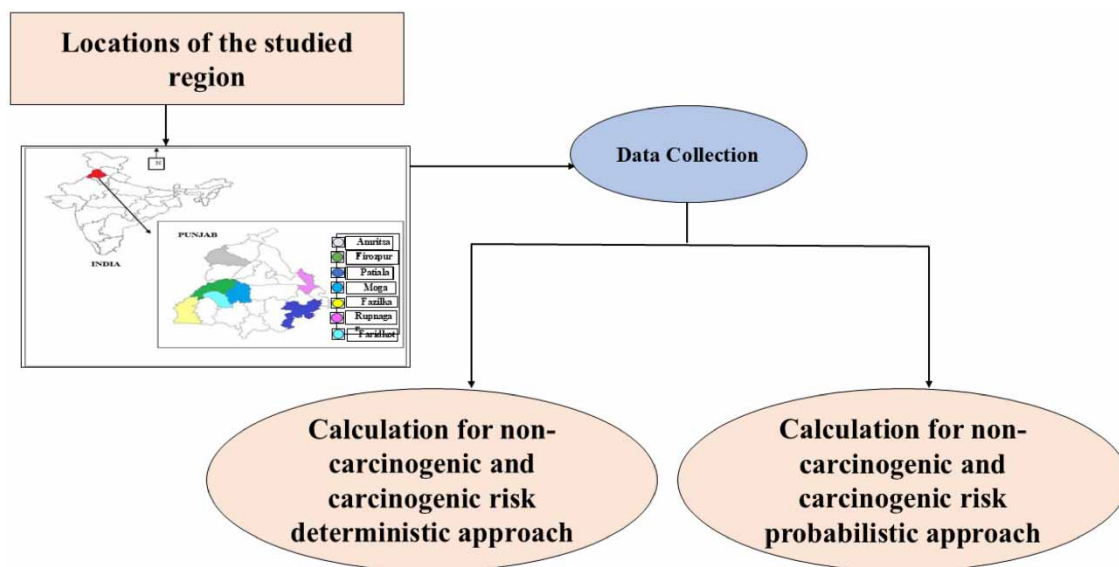
Groundwater contamination is a major environmental concern in many regions of India, including several districts of Punjab. In this study, a comparison is carried out between the deterministic and probabilistic approaches for calculating health hazard parameters due to arsenic contamination in groundwater in India. The probabilistic calculations are carried out through Monte-Carlo simulations to quantify the hazard index (HI) and carcinogenic risk. Sensitivity analysis is carried out to determine the contribution of input variables to the HI. From the deterministic and probabilistic approach, an HI greater than one is obtained in adults for all districts except Moga, whereas, for children, the HI is greater than 1 in all districts. This suggests a higher probability of risks, such as developmental problems and cardiovascular disorders. Excess lifetime carcinogenic risk, a measure of carcinogenic risk, is found above the US Environmental Protection Agency's recommended range of 1×10^{-6} to 1×10^{-4} in all districts. These values clearly indicate the long-term carcinogenic danger associated with arsenic exposure since they are far above the tolerable risk threshold of 1×10^{-4} . Thus, necessary mitigation measures should be taken, and routine groundwater monitoring should be performed.

Key words: arsenic, deterministic approach, groundwater, hazard index, Monte-Carlo simulations

HIGHLIGHTS

- Health impact.
- Risk assessment.
- Water contamination.
- Population health risk.
- Monte-Carlo simulations.

GRAPHICAL ABSTRACT



1. INTRODUCTION

Arsenic (As), characterized by its atomic number 33, is a widespread element that exists ubiquitously in the environment, seamlessly integrating into air, water, soil, and rocks (Ali *et al.* 2019). In developing countries, most of the general population consumes arsenic-contaminated drinking water several times higher than the World Health Organization's (WHO) acceptable limit of $10 \mu\text{g L}^{-1}$ (World Health Organization 1996; Hassan *et al.* 2009). According to numerous reports, over 100 million population globally have been affected by arsenic concentrations of greater than $10 \mu\text{g L}^{-1}$ (Rahman *et al.* 2005; Nriagu *et al.* 2007). The acceptable limit of arsenic in groundwater is set at $50 \mu\text{g L}^{-1}$ in many developing countries (Nickson *et al.* 2005), and 42 million people are exposed to more than $50 \mu\text{g L}^{-1}$ arsenic concentration in potable water. Due to arsenic's high toxicity, it can have adverse health effects even at low exposure levels, including dryness of the throat and mouth, dysphasia, profuse diarrhea, vomiting, melanosis, gastrointestinal distress, immune system suppression, superficial as well as internal cancers, and cardiovascular and neurological effects (Murphy *et al.* 1981; Wu *et al.* 1989; Chiou *et al.* 1995; Mandal *et al.* 1996; Knobeloch & Zierold 2006). It is a common metalloid prevalent in igneous and sedimentary rocks on the surface of the Earth's crust (Singh *et al.* 2010). In addition, human activities can discharge high contents of arsenic into the environment (Saha *et al.* 2017; Senila *et al.* 2017). There are four oxidation states of arsenic found in the environment: elemental arsenic (0), arsenate (+5), arsenite (+3), and arsenic (−3) (Ammann 2011). Arsenic (+5) exists in four different forms in aqueous solution, depending on the pH: H_3AsO_4 , H_2AsO_4^- , HAsO_4^{2-} , and AsO_4^{3-} . Arsenic (+3) exists in five different forms in an aqueous solution: H_4AsO_3^+ , H_3AsO_3 , H_2AsO_3^- , HAsO_3^- , and AsO_3^{3-} . The solubility of arsenic salts depends on pH, but elemental arsenic is insoluble in water (Edwards *et al.* 1998). Inorganic arsenic compounds contain trivalent and pentavalent arsenic; trivalent arsenic is more toxic and is four to ten times more soluble in water than pentavalent arsenic (Tuzen *et al.* 2009; Brahman *et al.* 2013). Analysis of the valency and speciation of soluble arsenic can significantly help in improving the preparation of an arsenic removal strategy. The most prevalent species of arsenic is arsenite, whereas arsenate is stable in oxygenated aquatic environments. Arsenic, in its elemental form, is redox-sensitive. Numerous geochemical parameters, including pH levels, reduction–oxidation processes, aquatic chemistry, distribution of other ionic species, and microbial activity, combine to determine its presence, mobility, distribution, and forms. Groundwater is more susceptible to arsenic contamination than surface water because anoxic conditions in subsurface environments increase arsenic's mobility (Shih 2005).

Groundwater in Argentina, Hungary, Mexico, and various regions of the United States is heavily polluted by arsenic (Thakur & Semil 2013). According to several studies (Mandal & Suzuki 2002; Smedley & Kinniburgh 2002; Agusa *et al.* 2004; Stanger *et al.* 2005; Sengupta *et al.* 2006; Yadav *et al.* 2014), there is a concerning danger of arsenic pollution in South-east Asian nations, including India, Bangladesh, China, Nepal, and Vietnam. According to Nriagu *et al.* (2002), thermal springs and mining operations are important sources of arsenic release into groundwater under oxidizing circumstances.

The highest concentration of arsenic in Canada has been found to be roughly 100 mg L^{-1} in groundwater (Nriagu *et al.* 2002). Australia has underground water significantly contaminated with concentrations of arsenic in groundwater that have been reported as high as 300 mg L^{-1} (Nriagu *et al.* 2002). Mining operations, oxidation of minerals containing sulfides, and volcanic eruptions have all been identified as major causes of arsenic release into groundwater (Boyle *et al.* 1998). In developing nations including India, groundwater contamination with arsenic is one of the most significant and hazardous environmental problems (Environmental Protection Agency 2001; Prakash & Verma 2021). Kumar *et al.* (2016) and Shaikh *et al.* (2016) reported $84.28 \mu\text{g L}^{-1}$ arsenic concentration in the groundwater of Bihar state. In Jharkhand, Alam *et al.* (Singh 2004; Alam *et al.* 2016) reported the arsenic concentration higher than $10 \mu\text{g L}^{-1}$. Sharma *et al.* (2013) reported that arsenic contamination is greater than $10 \mu\text{g L}^{-1}$ in groundwater in the eastern part of India. In the Indian state of Punjab, groundwater is an essential source of drinking water and is used in all aspects of human life, including industries, agriculture, and residences (Gupta 2009). About 95% of the people in Punjab get their drinking water from groundwater (Hundal *et al.* 2009).

The primary source of arsenic in groundwater is the dissolution of minerals from weathered rocks and soils. In 1967, Punjab had just about 50,000 tube wells, but in 2008, this number surpassed one million (Hundal *et al.* 2009). In the previous three decades, groundwater extraction has increased by 200 times in this state.

In this paper, a comparative study of both approaches for the risk assessment due to contamination in groundwater of Moga, Faridkot, Fazilka, Patiala, Ferozepur, Rupnagar, and Amritsar districts of Punjab is performed. These regions are chosen based on their geographic locations and the increasing rate of cancer-related mortality due to the intake of water of inferior quality. The dependance of health risk parameters on the input variables is determined using the sensitivity analysis. The concentration of arsenic in these districts is reported in the literature (Central Ground Water Board 2013; Kumar *et al.* 2017; Sidhu *et al.* 2018; Virk 2019). In these studies, health risk assessment due to arsenic contamination is not carried out.

In the previous studies, the deterministic approach was used for the human health risk assessment due to arsenic contamination in groundwater. This method considers the site-specific data to estimate the risk of non-carcinogenic adverse health effects in adults and children, but it does not address the inherent uncertainty of input parameters. There are various variables in calculations of health risk for which assuming a point distribution might not incorporate the inherent variability in the input parameters used for health risk assessment. In order to avoid the uncertainty associated with the deterministic methodology, Monte-Carlo simulations were carried out for the non-carcinogenic as well as carcinogenic risk assessment in adults and children due to the presence of arsenic in groundwater.

2. MATERIALS AND METHOD

2.1. Study area

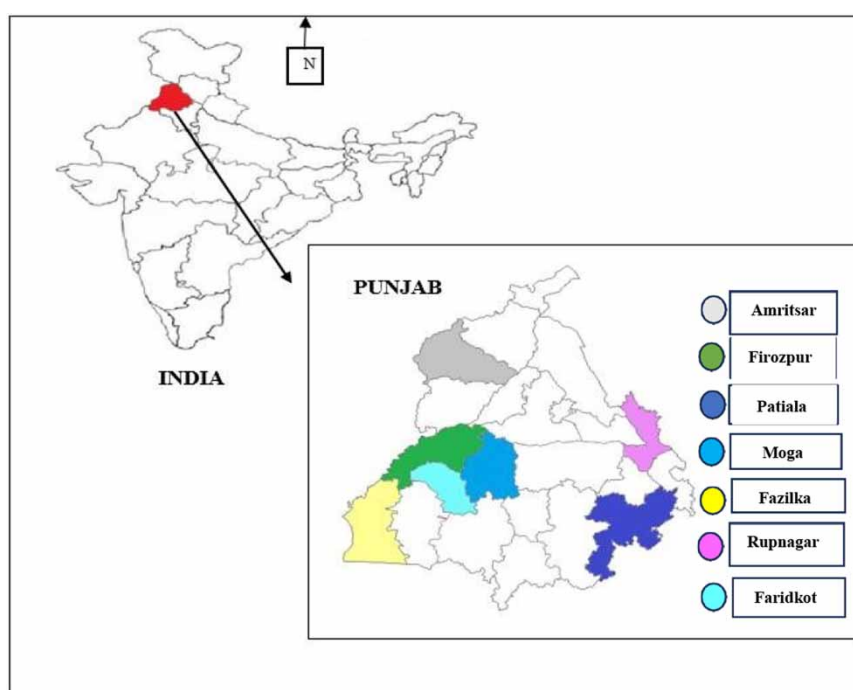
Punjab is a state in India from 29.30° N to 32.32° N and 73.55° E to 76.50° E . The occupied area of this state is $50,476 \text{ km}^2$. Unconsolidated alluvium covers almost an entire Punjab area. The climate of Punjab affects its hydrogeology. Groundwater availability and recharge are affected by the state's varying rainfall patterns. The rivers Beas, Ravi, and Sutlej, and the seasonal Ghaggar flow through the state, and the average annual precipitation ranges from 400 to 1,300 mm. A comprehensive network of canals supports the state's irrigation system. Geographically, Punjab state is segregated into three different regions: Majha, which lies between the rivers Ravi and Beas; Doaba, which lies between the rivers Sutlej and Beas; and Malwa, which is located south of the Sutlej. The mean value of arsenic concentration in groundwater in the chosen districts of Punjab is listed in Table 1. Moga, Faridkot, Fazilka, Patiala, and Ferozepur districts are in the Malwa region, Rupnagar is located in the Doaba region and Amritsar is in the Majha region of Punjab. The locations of the studied districts are shown in Figure 1.

2.2. Sample collection and technique

In order to evaluate the arsenic level in groundwater, Mandeep Sindhu *et al.* collected water samples using hand pumps from several locations in the Moga and Faridkot districts of Punjab (Sidhu *et al.* 2018). The author collected a total of 40 groundwater samples from each district using a random sampling approach. For a comprehensive coverage of the region, these samples were obtained from several hand pumps. After collection, 1 milliliter (mL) of hydrochloric acid was added to each sample for preservation after measuring the pH of the sample. Each water sample was tested by pouring 100 mL into a glass beaker, and 2 mL of concentrated nitric acid (HNO_3) was added. The combination was brought to a temperature

Table 1 | Mean value of arsenic concentration in groundwater in the studied regions (Central Ground Water Board 2013; Kumar *et al.* 2017; Sidhu *et al.* 2018; Virk 2019)

Districts	Mean value of arsenic concentration in groundwater (mg L ⁻¹)
Moga	0.0043
Faridkot	0.017
Fazilka	0.037
Patiala	0.021
Ferozepur	0.027
Rupnagar	0.035
Amritsar	0.083

**Figure 1** | Location of districts of Punjab for which health risk assessment is carried out.

of 90–95 °C in a water bath, covered with a watch glass, and left there until the liquid evaporated to the extent of 20 mL. After the solution cooled, any insoluble minerals or silicates in the sample were removed by filtering it using the Whatman filter paper (grade I). The samples were prepared for analysis after adding reagent water to get the volume back up to 100 mL. After that, the arsenic concentration of the water was determined using the inductive coupled argon plasma–atomic emission spectroscopy (ICAP–AES). To analyze the level of arsenic contamination in groundwater samples from the Fazilka, the digested filtrates were produced using a diethylenetriamine penta acetic acid extraction procedure (Virk 2019). After being produced, the quantity of arsenic in these filtrates was determined by using an inductively coupled plasma–AES (Thermoelectric ICAP-6300 model). Groundwater samples were taken using a random sampling technique in Patiala, Ferozepur, Rupnagar, and Amritsar. To avoid contamination, HNO₃ was used to clean the polyethylene plastic bottles before the sample collection. The bottles were then given a final washing with distilled water prior to the collection of 1,000 mL of flowing groundwater. Three milliliters of pure nitric acid of analytical grade were added to each sample. The arsenic content of the groundwater was measured using a graphite furnace–atomic absorption spectrometer, more precisely the 240 FS AA model from Agilent Technologies, Santa Clara, California, USA. Before analyzing the samples, the spectrometer was carefully

calibrated, and a blank run was performed to ensure accuracy in the readings (Central Ground Water Board). The measured concentration of arsenic in groundwater in all districts is above the permissible limit of 0.01 mg/L.

3. HUMAN HEALTH RISK ASSESSMENT METHODS

3.1. Non-carcinogenic risk assessment

Human health risks are quantified using the United States Environmental Protection Agency's (USEPA) approved health risk model. A quantitative evaluation of the impact of environmental pollution on human health is known as a health risk assessment. In the present study, the non-cancer risk due to arsenic contamination in groundwater is calculated by evaluating the hazard quotient (HQ) considering ingestion and dermal routes.

The average daily dose for ingestion and dermal is calculated for adults and children by using the following equations, respectively.

$$ADD_{ing} = \frac{C \times IR \times EF \times ED}{BW \times AT} \quad (1)$$

$$ADD_{dermal} = \frac{C \times K_p \times EF \times ED \times ET \times SA \times CF}{BW \times AT} \quad (2)$$

where ADD_{ing} is the average daily dose for ingestion ($\text{mg kg}^{-1} \text{ day}^{-1}$), ADD_{dermal} is the average daily dose due to dermal contact with ($\text{mg kg}^{-1} \text{ day}^{-1}$), C is the concentration of arsenic in the water sample (mg L^{-1}), IR is the daily ingestion rate of water (L day^{-1}), EF is the exposure frequency (day year^{-1}), ED is the exposure duration (years), BW is the body weight (kg), AT is the average time (days), SA is the exposed skin surface area (m^2), CF is the conversion factor (L-m/cm-m^3). For the deterministic and probabilistic calculations, the values of these parameters are listed in Table 2, respectively.

Table 2 | The model parameters assumed in the deterministic approach for the calculation of HI in adults and children (USEPA 1989, 1996, 2004, 2007; Sharma *et al.* 2013)

Deterministic approach		Probabilistic approach	
Input parameters	Values (units)	Distribution type	Values (units)
Concentration of arsenic (C)	Mean (mg/L)	Lognormal	Metal specific (mg L^{-1})
IR adult	2 (L day^{-1})	Lognormal	1.26 ± 0.66 (L day^{-1})
IR children	1.8 (L day^{-1})	Lognormal	1.26 ± 0.66 (L day^{-1})
EF	365 (day year^{-1})	Triangular	345 (180–365) (day year^{-1})
ED adult	70 (years)	Point	70 (years)
ED children	6 (years)	Point	6 (years)
Exposure time (ET)	0.58 (h day^{-1})	Triangular	0.20 (0.13–0.33) (h day^{-1})
BW adult	70 (kg)	Lognormal	77.1 ± 31.5 (kg)
BW children	15 (kg)	Triangular	26.1 (6.5–15) (kg)
AT adult	25,550 (days)	Point	25,550 (days)
AT children	2,190 (days)	Point	2,190 (days)
Permeability constant (K_p)	0.001 (cm h^{-1})	Point	0.001 (cm h^{-1})
Exposed skin SA adult	1.8 (m^2)	Lognormal	1.42 ± 0.31 (m^2)
Exposed skin SA children	0.6 (m^2)	Triangular	0.6800 ± 0.600 (m^2)
CF	10	Point	10 (L-m) ($\text{m}^3\text{-cm}$)
Reference dose (RfD_{ing}) ingestion	0.0003 ($\text{mg kg}^{-1} \text{ day}^{-1}$)	Point	0.0003 ($\text{mg kg}^{-1} \text{ day}^{-1}$)
Reference dose (RfD_{dermal}) dermal	0.000285 ($\text{mg kg}^{-1} \text{ day}^{-1}$)	Point	0.000285 ($\text{mg kg}^{-1} \text{ day}^{-1}$)

The HQ for ingestion and dermal is calculated from the following equations, respectively.

$$HQ_{\text{ing}} = \frac{ADD_{\text{ing}}}{RfD_{\text{ing}}} \quad (3)$$

$$HQ_{\text{dermal}} = \frac{ADD_{\text{dermal}}}{RfD_{\text{dermal}}} \quad (4)$$

where RfD_{ing} and RfD_{dermal} are the reference doses ($\text{mg kg}^{-1} \text{ day}^{-1}$) of arsenic for ingestion and dermal, respectively. Reference doses are limited up to which intake of chemical doses does not pose a significant chemical risk. The non-cancer risk associated with arsenic ingestion and dermal contact with arsenic is determined through the hazard index (HI). The HI is calculated by the sum of the HQ of ingestion and dermal exposure, expressed as follows:

$$HI = HQ_{\text{ing}} + HQ_{\text{dermal}} \quad (5)$$

An HI of more than one indicates that there may be adverse effects on human health, emphasizing the need for careful consideration and possible mitigating actions.

3.2. Excess lifetime carcinogenic risk

The excess lifetime carcinogenic risk (ELCR) has been determined by using the average daily dose and cancer slope factor (CSF). The cancer-causing threat posed by oral ingestion of carcinogens is calculated from the following equation.

$$ELCR = ADD_{\text{ing}} \times CSF \quad (6)$$

CSF is a heavy metal-dependent factor based on actual studies demonstrating the adverse health effects of specific carcinogenic pollutants. According to the USEPA recommendation, for arsenic, the CSF is $1.5 \text{ mg kg}^{-1} \text{ day}^{-1}$ for the ingestion pathway.

The permissible range of carcinogenic risk is between 10^{-6} and 10^{-4} , whereas the maximum acceptable level recommended by the WHO is 10^{-4} .

3.3. Deterministic approach

A deterministic approach is used when limited data are available or when the system being studied is well understood. In the deterministic method, the health risk is assessed from the known parameters, and it focuses on assessing the impact of single-value risk. This method considers the site-specific data to estimate the risk of carcinogenic and non-carcinogenic elements on human health in adults and children, but there is inherent uncertainty and variability in the parameters used in the calculations. To calculate a single-point estimate of risk, the various parameters used in calculations are listed in [Table 2](#).

3.4. Probabilistic approach (Monte-Carlo simulations)

Evaluation of potential risks to human health is inherently ambiguous due to individual human characteristics and environmental variables. A more realistic approach (Monte-Carlo simulations) to risk calculation can be achieved by addressing issues of variability and uncertainty. It is used to assess the risks of carcinogenic and non-carcinogenic risks in adults and children due to direct exposure to contaminated water via ingestion and dermal pathways. To accomplish this, Monte-Carlo simulations are used to estimate the non-carcinogenic and carcinogenic risks based on their probability distributions of various parameters. In this method, input parameters are defined as probability distributions resulting in the average daily dose and the HI. The value of each parameter is randomly chosen from the probability distribution function as given in [Table 2](#). To test the reliability of the results, various independent runs were conducted with 10,000 iterations to estimate the non-cancer risk. The results of non-carcinogenic and carcinogenic risks are given in [Table 4](#). The non-carcinogenic risk falls within an acceptable range if their HI value is less than 1.

4. RESULTS AND DISCUSSION

4.1. Non-carcinogenic risk assessment

The analysis of groundwater pollution by arsenic in Punjab's districts shows significant regional variation. Amritsar has the highest arsenic content, whereas Moga has the lowest. All districts except Moga have arsenic levels over the WHO-recommended 0.01 mg/L threshold. A previous study on groundwater contamination due to arsenic exposure in the area affirms these results (Table 3). The deterministic approach is used to calculate the HI for adults and children in the seven districts (Table 4). The findings demonstrate that children are more vulnerable than adults. In Moga, Faridkot, Fazilka, Patiala, Ferozepur, Rupnagar, and Amritsar, the corresponding HI values for adults are 0.41, 1.62, 3.54, 2.01, 2.58, 3.35, and 7.94. The equivalent values for children are much higher: 1.72, 6.83, 14.87, 8.44, 10.85, 14.07, and 33.26. According to the deterministic analysis, children are more at risk for non-carcinogenic diseases due to arsenic exposure; in all districts,

Table 3 | Arsenic concentration in groundwater of the study area to India and world

Study area	Concentration (mg L ⁻¹)	Reference
Moga	0.0043	Present study
Faridkot	0.017	
Fazilka	0.037	
Patiala	0.021	
Ferozepur	0.027	
Rupnagar	0.035	
Amritsar	0.083	
Ropar	0.011	Mandal <i>et al.</i> (1996)
West Bengal	0.025	Yadav <i>et al.</i> (2020)
Fatehabad	0.050 to 4.531	Vige <i>et al.</i> (2023)
Jharkhand	0.082	Giri & Singh (2015)
Iran	0.012	Barzegar <i>et al.</i> (2019)
China	0.002–0.188	Zhaoyong <i>et al.</i> (2018)
Vietnam	0.211–0.348	Nguyen <i>et al.</i> (2009)
Pakistan	0.107	Mushtaq <i>et al.</i> (2020)
Srilanka	0.34	Bandara <i>et al.</i> (2018)
Taiwan	0.975	Maity <i>et al.</i> (2019)
Qatar	0.15	Da'ana <i>et al.</i> (2021)
Mexico	0.073	Ruj <i>et al.</i> (2021)
USA	0.150	Tsuji <i>et al.</i> (2019)

Table 4 | From the deterministic approach, the HI and carcinogenic risk values for adults and children

Districts	HI (adults)	HI (children)	Carcinogenic risk (adults)	Carcinogenic risk (children)
Moga	0.41	1.72	1.84×10^{-4}	7.74×10^{-4}
Faridkot	1.62	6.83	7.29×10^{-4}	3.06×10^{-3}
Fazilka	3.54	14.87	1.59×10^{-3}	6.66×10^{-3}
Patiala	2.01	8.44	9.01×10^{-4}	3.78×10^{-3}
Ferozepur	2.58	10.85	1.16×10^{-3}	4.80×10^{-3}
Rupnagar	3.35	14.07	1.50×10^{-3}	6.30×10^{-3}
Amritsar	7.94	33.26	3.50×10^{-3}	1.40×10^{-2}

including Moga, the HI value is more significant than 1 for children. On the other hand, adults in Moga have the HI values of less than 1, indicating no risk. All other districts, however, surpass the threshold, with Amritsar's adults being at the highest risk. The elevated HI values in children highlight their heightened vulnerability, especially in areas like Fazilka, Ferozepur, and Amritsar, where the HI levels surpass 10, indicating severe health issues such as cardiovascular disease, skin lesions, and cancer.

The probabilistic approach used the 95th percentile values to consider variability and avoid overestimation, and the HI values thus calculated are as listed in Table 5. The findings demonstrate that non-carcinogenic hazards are significantly high. In Moga, Faridkot, Fazilka, Patiala, Ferozepur, Rupnagar, and Amritsar, the HI values at the 95th percentile for adults are 0.78, 2.70, 4.78, 2.76, 3.45, 6.97, and 10.32, respectively. The corresponding values for children are 3.53, 11.42, 20.93, 11.80, 14.47, 30.44, and 44.73, respectively. This supports the idea that exposure to arsenic presents serious health hazards, particularly for children. Additionally, district-specific risk exposure patterns are shown using the probabilistic estimation. 94% of adults and 98% of children in Amritsar have an HI of more than 1, and more than 87% of people have an HI of 2, which indicates possible severe health problems. There are also alarming trends in other areas, such as Rupnagar, Fazilka, and Ferozepur, especially among children, where more than 70% of the child population has HI values of more than 1. The detailed results showing various percentiles for all districts from Monte-Carlo simulations are shown from Figures 2 to 8. These results demonstrate the concerning health effects of groundwater consumption-related arsenic exposure, particularly for children, who are disproportionately more vulnerable in all districts except Moga. A large population may have negative

Table 5 | From the probabilistic approach, HI, and carcinogenic risk values for adults and children

Districts	95th percentile of the HI (adults)	95th percentile of cancer risk (adult)	95th percentile of the HI (children)	95th percentile of cancer risk (children)
Moga	0.78	3.51×10^{-4}	3.53	1.58×10^{-3}
Faridkot	2.70	1.21×10^{-3}	11.42	5.14×10^{-3}
Fazilka	4.78	2.15×10^{-3}	20.93	9.41×10^{-3}
Patiala	2.76	1.24×10^{-3}	11.80	5.30×10^{-3}
Ferozepur	3.45	1.46×10^{-3}	14.47	6.51×10^{-3}
Rupnagar	6.97	3.10×10^{-3}	30.44	1.36×10^{-2}
Amritsar	10.32	4.60×10^{-3}	44.73	2.01×10^{-2}

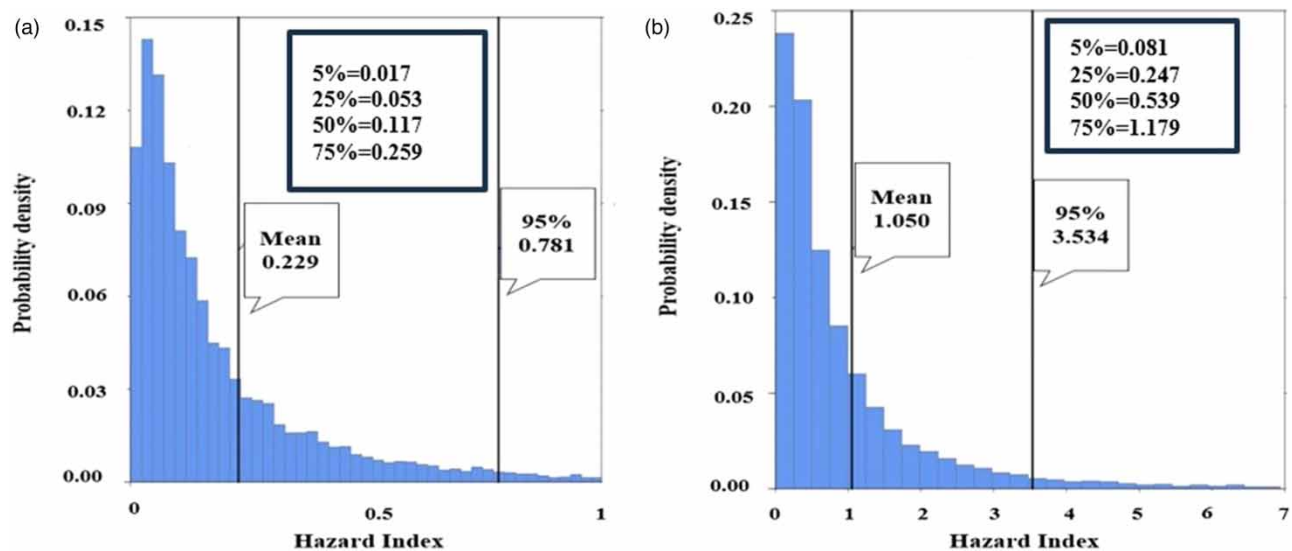


Figure 2 | Probabilistic estimates of the HI for (a) adults and (b) children in Moga district.

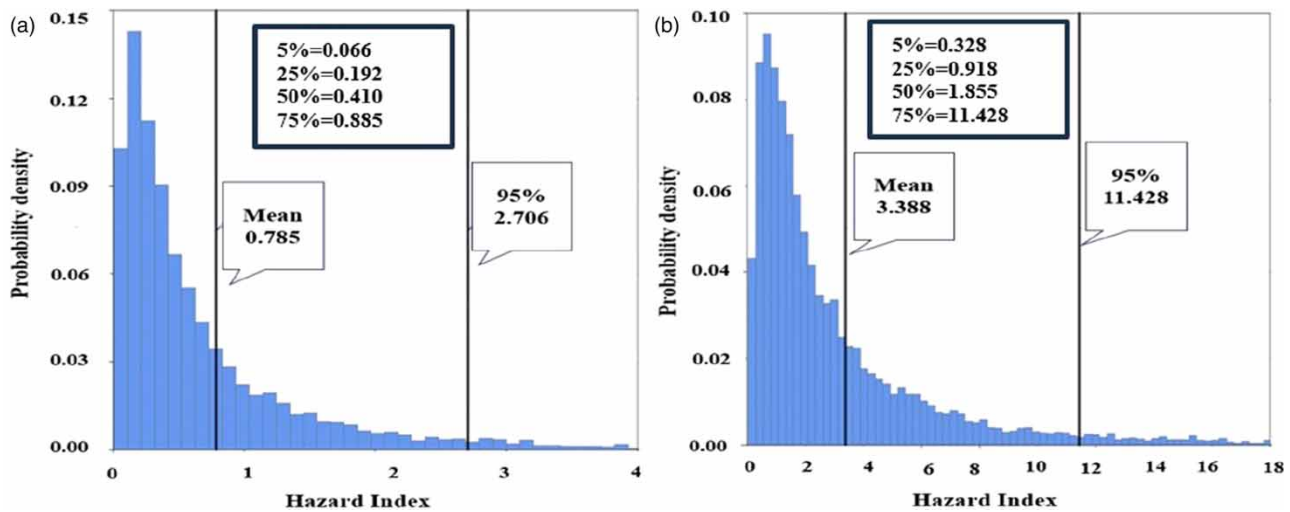


Figure 3 | Probabilistic estimates of the HI for (a) adults and (b) children in Faridkot district.

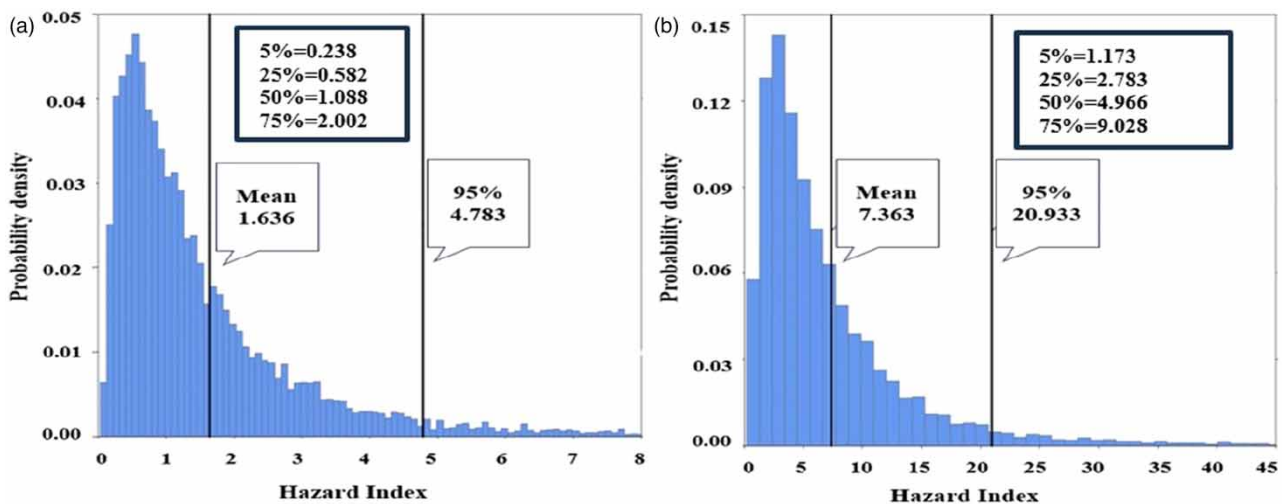


Figure 4 | Probabilistic estimates of the HI for (a) adults and (b) children in Fazilka district.

health impacts in areas with the greatest arsenic contamination such as Amritsar. This highlights how urgently targeted actions are needed to reduce arsenic exposure, especially in areas where most children have the HI values much higher than 2.

4.2. Carcinogenic risk assessment

Long-term exposure to contaminated groundwater raises the risk of acquiring cancer since arsenic is a proven carcinogen. The ELCR was assessed using probabilistic and deterministic approaches in each of the studied districts. According to the deterministic analysis, the cancer risk for adults in the Moga, Faridkot, Fazilka, Patiala, Ferozepur, Rupnagar, and Amritsar are 1.84×10^{-4} , 7.29×10^{-4} , 1.59×10^{-3} , 9.01×10^{-4} , 1.16×10^{-3} , 1.50×10^{-3} , and 3.50×10^{-3} , respectively. The ELCR for children are significantly higher: 7.74×10^{-4} , 3.06×10^{-3} , 6.66×10^{-3} , 3.78×10^{-3} , 4.80×10^{-3} , 6.30×10^{-3} , and 1.40×10^{-2} , in the same order. Particularly in Fazilka, Rupnagar, and Amritsar, the values in several districts above the WHO's threshold limit of tolerable cancer risk (1×10^{-4}). The cancer risk for adults in Moga, Faridkot, Fazilka, Patiala, Ferozepur, Rupnagar, and Amritsar is estimated using the 95th percentile probabilistic approach to be 3.51×10^{-4} , 1.21×10^{-3} , 2.15×10^{-3} , 1.24×10^{-3} , 1.46×10^{-3} , 3.10×10^{-3} , and 4.60×10^{-3} , respectively. 1.58×10^{-3} , 5.14×10^{-3} , 9.41×10^{-3} , 5.30×10^{-3} , 6.51×10^{-3} ,

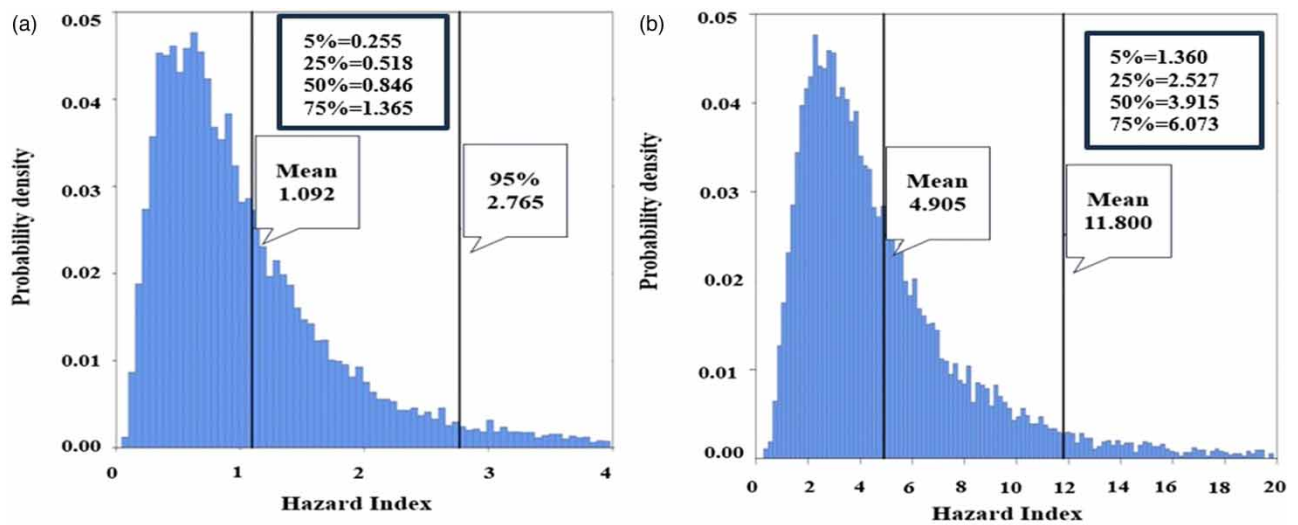


Figure 5 | Probabilistic estimates of the HI for (a) adults and (b) children in Patiala district.

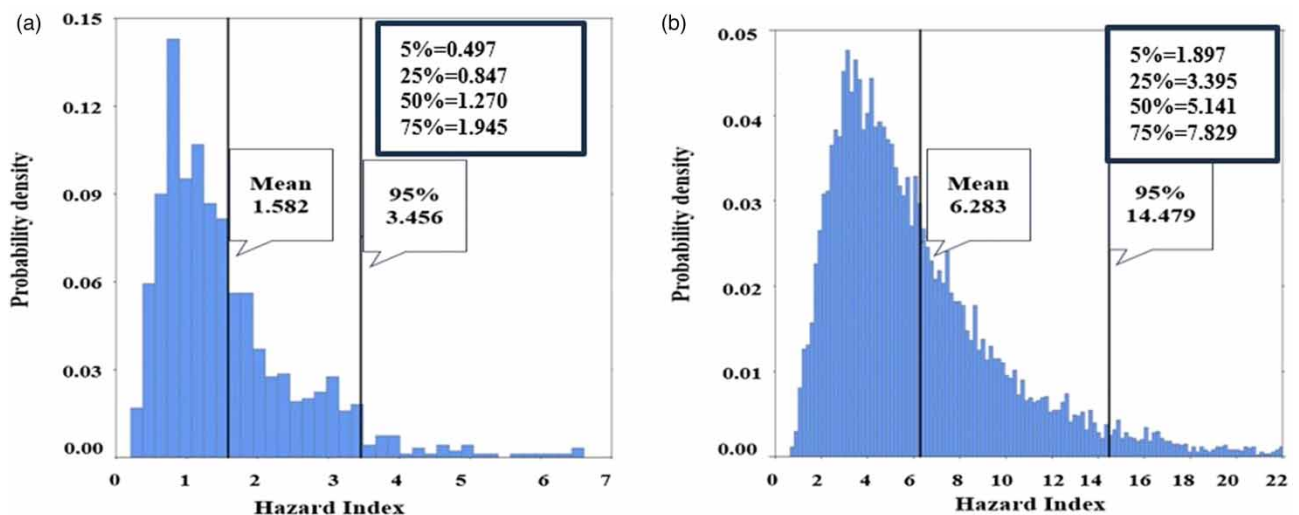


Figure 6 | Probabilistic estimates of the HI for (a) adults and (b) children in Ferozepur district.

1.36×10^{-2} , and 2.01×10^{-2} are the similar values for children. More than 10 children out of every 1,000 might get cancer from arsenic exposure in Rupnagar and Amritsar, where the cancer risk for children is more than 1.36×10^{-2} and 2.01×10^{-2} , respectively. These findings highlight the seriousness of the carcinogenic hazards associated with arsenic exposure, especially for children. Long-term exposure to tainted groundwater may raise the incidence of cancer in these areas, which has serious public health ramifications. Focused remediation techniques and public health initiatives are essential to address this issue effectively.

4.3. Limitations and future considerations

While this study provides critical insights into the health risks posed by arsenic contamination in groundwater, limitations are acknowledged. This study only looks at arsenic, leaving out other potentially dangerous pollutants, including nickel (Ni), lead (Pb), and iron (Fe), which are often co-contaminants in groundwater. A more thorough examination that considers a number of pollutants should be included in future research. To increase the accuracy of risk evaluations, long-term studies with more detailed exposure data should be performed. Furthermore, a more comprehensive knowledge of the health effects of arsenic

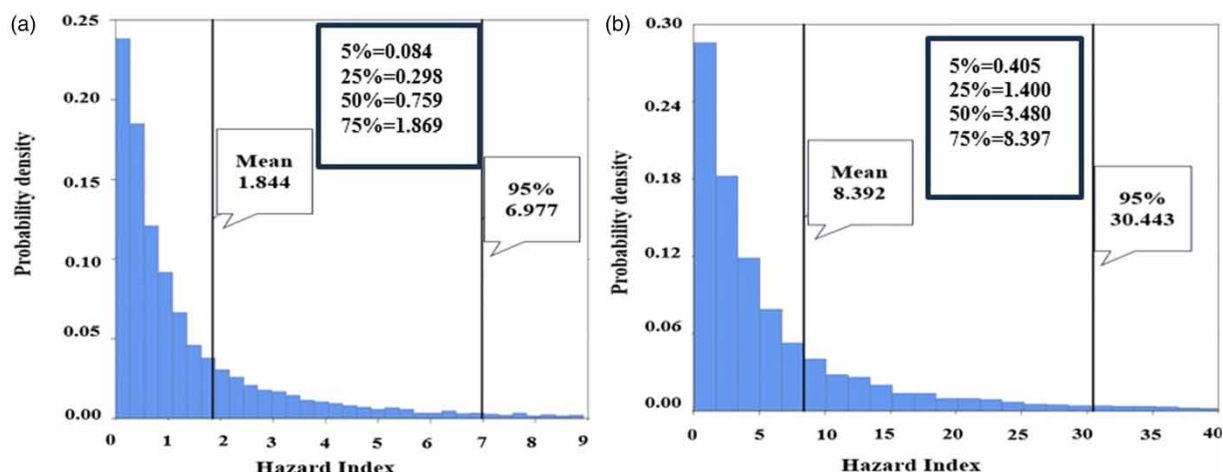


Figure 7 | Probabilistic estimates of the HI for (a) adults and (b) children in Rupnagar district.

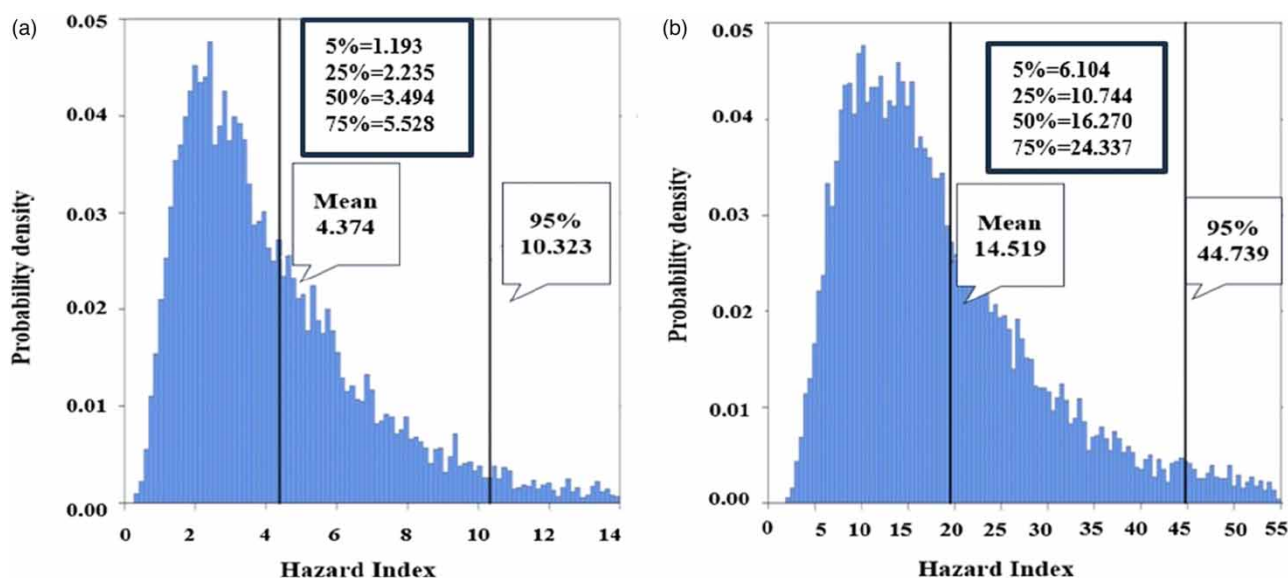


Figure 8 | Probabilistic estimates of the HI for (a) adults and (b) children in Amritsar district.

exposure in Punjab could be achieved by diversifying the research to include other areas and using data from continuous monitoring.

4.4. Sensitivity analysis

Sensitivity analysis is used to determine the predictor factors' contribution to the prediction. It serves as an example of the most crucial elements that affect the assessment's outcome. The study included a sensitivity analysis of the non-carcinogenic risk simulation parameters. The findings are plotted as tornado plots that display the Spearman rank-order correlation coefficients on a decimal scale. The sensitivity analysis of variables in calculating HI for two exposed groups, adults and children, in the different given districts is shown in Figures 9–15. The sensitivity analysis findings indicated that the arsenic concentration parameter had the most impact on increasing sensitivity in the two exposed groups under study. Consequently, mitigation strategies should focus on decreasing arsenic concentration in drinking water, which will probably reduce health risks. The concentration of arsenic in the groundwater is the most significant variable for these two groups of the population,

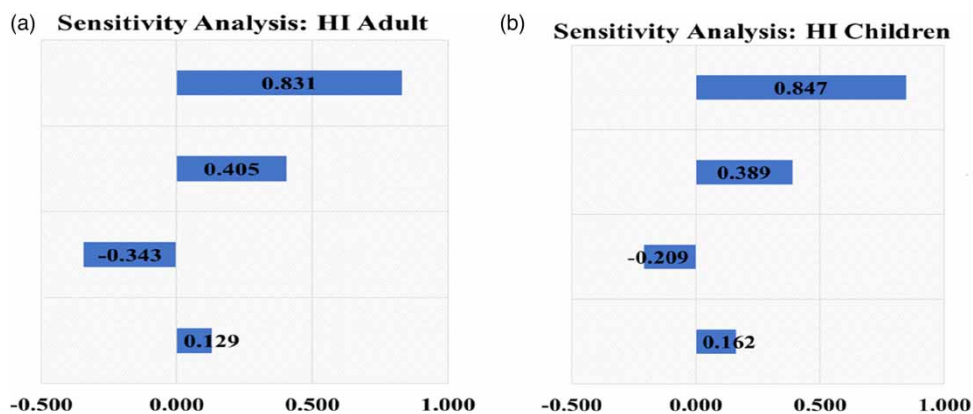


Figure 9 | Sensitivity analysis of the non-carcinogenic risk of (a) adults and (b) children in the Moga district.

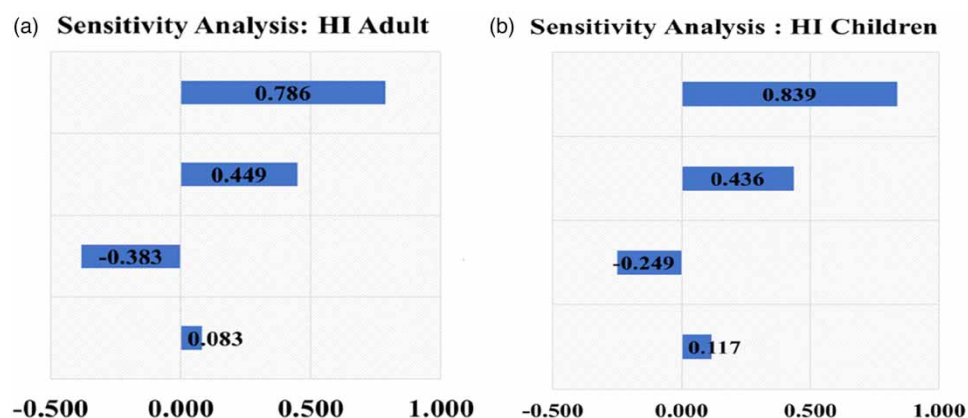


Figure 10 | Sensitivity analysis of the non-carcinogenic risk of (a) adults and (b) children in the Faridkot district.

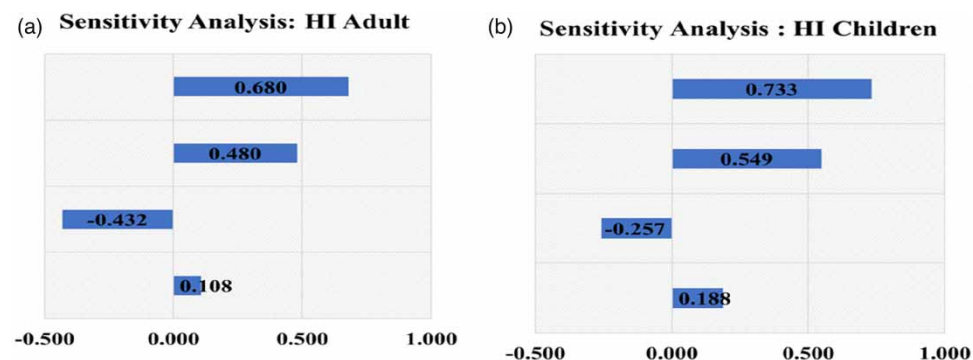


Figure 11 | Sensitivity analysis of the non-carcinogenic risk of (a) adults and (b) children in the Fazilka district.

with the Spearman rank-order correlation coefficient being 0.831 and 0.847 in Moga, 0.786 and 0.839 in Faridkot, 0.680 and 0.733 in Fazilka, 0.647 and 0.722 in Patiala, 0.693 and 0.788 in Ferozepur, 0.871 and 0.895 in Rupnagar, and 0.739 and 0.812 in Amritsar for adult and child populations, respectively. The health risk is found to be least sensitive to ED. As expected, BW has a negative correlation with health risks.

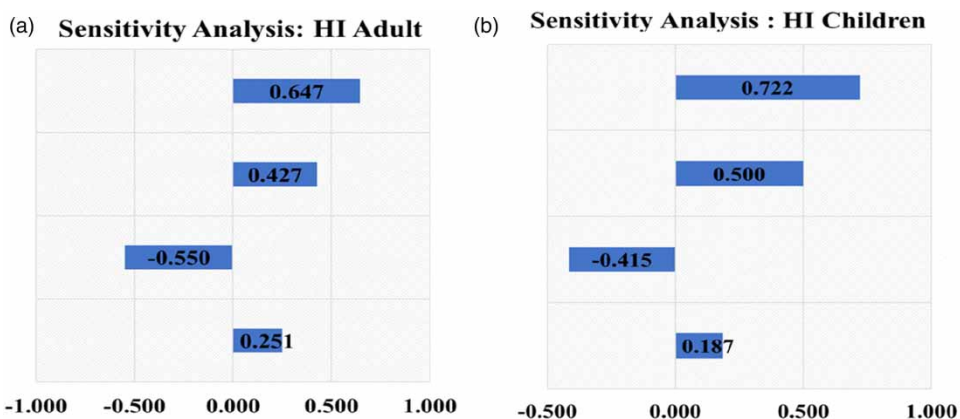


Figure 12 | Sensitivity analysis of the non-carcinogenic risk of (a) adults and (b) children in the Patiala district.

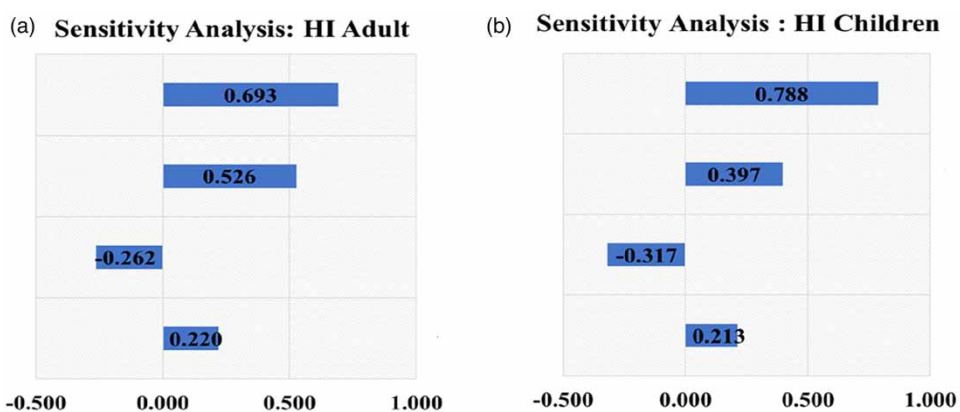


Figure 13 | Sensitivity analysis of the non-carcinogenic risk of (a) adults and (b) children in the Ferozepur district.

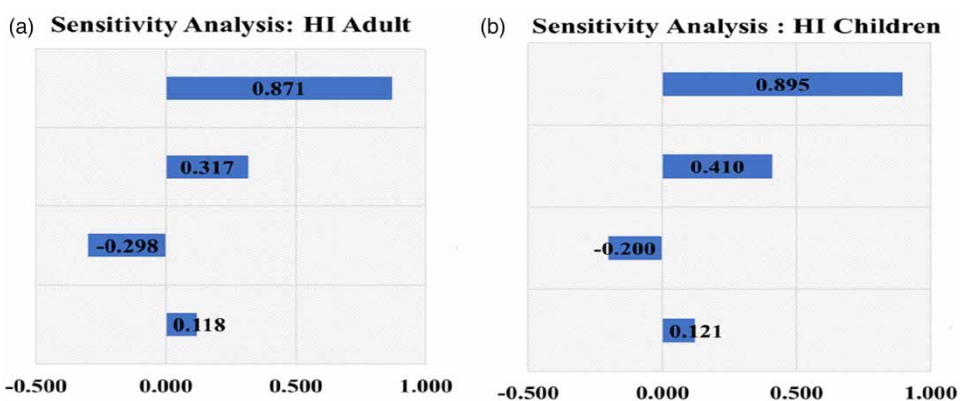


Figure 14 | Sensitivity analysis of the non-carcinogenic risk of (a) adults and (b) children in the Rupnagar district.

5. CONCLUSION

This study highlights the significant health concerns associated with arsenic-contaminated groundwater in Punjab's seven districts, where the majority of the population, especially children, is at higher risk for both non-carcinogenic and carcinogenic

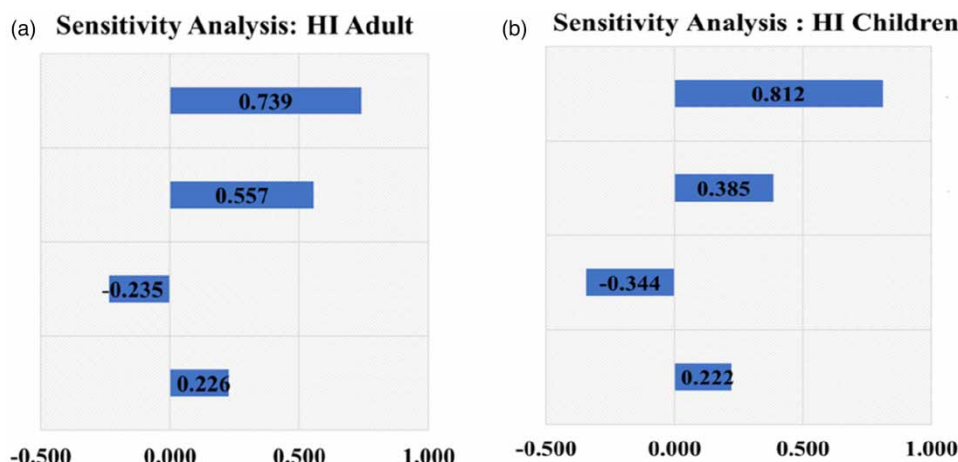


Figure 15 | Sensitivity analysis of the non-carcinogenic risk of (a) adults and (b) children in the Amritsar district.

diseases. For children in every district, the HI values for non-carcinogenic hazards are more than one, suggesting possible health problems such as neurological damage, cardiovascular illnesses, and developmental delays. Moreover, high ELCR indicates substantial long-term cancer risks, with certain districts such as Amritsar, Rupnagar, and Ferozepur exhibiting the greatest risk of developing cancer as a result of arsenic exposure. Monte-Carlo simulations have revealed the variability in exposure and risk, helping to provide a more nuanced risk assessment that highlights the disproportionate vulnerability of children. These results highlight how urgent arsenic mitigation measures, including better water treatment, public health campaigns, and continuous groundwater quality monitoring, are needed. In order to protect the health of the impacted communities and reduce exposure levels, particularly for pediatric populations, immediate action is essential. The study's findings provide authorities with a solid basis on which to build their priorities for reducing arsenic and enhancing public health in Punjab's impacted areas.

DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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